System Verification and Validation Plan for Software Engineering

Team 20, Aeolus Software
Andy Huynh
Andreas Sideris
Humna Saeed
Christopher Ludwig
Owen Johnson

November 1, 2026

Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

[The intention of the VnV plan is to increase confidence in the software. However, this does not mean listing every verification and validation technique that has ever been devised. The VnV plan should also be a **feasible** plan. Execution of the plan should be possible with the time and team available. If the full plan cannot be completed during the time available, it can either be modified to "fake it", or a better solution is to add a section describing what work has been completed and what work is still planned for the future. —SS]

[The VnV plan is typically started after the requirements stage, but before the design stage. This means that the sections related to unit testing cannot initially be completed. The sections will be filled in after the design stage is complete. the final version of the VnV plan should have all sections filled in. —SS]

Contents

1 Symbols, Abbreviations, and Acronyms **iv**

2 General Information **1**
- 2.1 Summary . . . 1
- 2.2 Objectives . . . 1
- 2.3 Custom Documents . . . 1
- 2.4 Relevant Documentation . . . 2

3 Plan **2**
- 3.1 Verification and Validation Team . . . 2
- 3.2 SRS Verification . . . 2
- 3.3 Design Verification . . . 3
- 3.4 Verification and Validation Plan Verification . . . 3
- 3.5 Implementation Verification . . . 3
- 3.6 Automated Testing and Verification Tools . . . 3
- 3.7 Software Validation . . . 4

4 System Tests **4**
- 4.1 Tests for Functional Requirements . . . 4
  - 4.1.1 Area of Testing1 . . . 4
  - 4.1.2 User Interface Tests for Functional Requirements . . . 5
- 4.2 Tests for Nonfunctional Requirements . . . 6
  - 4.2.1 Area of Testing1 . . . 6
  - 4.2.2 Area of Testing2 . . . 7
- 4.3 Traceability Between Test Cases and Requirements . . . 7

5 Unit Test Description **7**

6 Appendix **8**
- 6.1 Symbolic Parameters . . . 8
- 6.2 Usability Survey Questions? . . . 8

7 Appendix — Generative AI Use Disclosure **9**
- 7.1 AI Tools Used . . . 9
- 7.2 How and Where Used . . . 9
- 7.3 Verification of AI-Generated Content . . . 9

8 Appendix — Reflection **10**

List of Tables

1 RealmForge System Test Requirements Traceability Summary 7
2 Symbolic Constants Used in the Verification and Validation Plan 8

List of Figures

[Remove this section if it isn't needed —SS]

1 Symbols, Abbreviations, and Acronyms

symbol	description
T	Test

[symbols, abbreviations, or acronyms — you can simply reference the SRS [1] tables, if appropriate —SS]

[Remove this section if it isn't needed —SS]

This document ... [provide an introductory blurb and roadmap of the Verification and Validation plan —SS]

2 General Information

2.1 Summary

[Say what software is being tested. Give its name and a brief overview of its general functions. —SS]

2.2 Objectives

[State what is intended to be accomplished. The objective will be around the qualities that are most important for your project. You might have something like: "build confidence in the software correctness," "demonstrate adequate usability." etc. You won't list all of the qualities, just those that are most important. —SS]

[You should also list the objectives that are out of scope. You don't have the resources to do everything, so what will you be leaving out. For instance, if you are not going to verify the quality of usability, state this. It is also worthwhile to justify why the objectives are left out. —SS]

[The objectives are important because they highlight that you are aware of limitations in your resources for verification and validation. You can't do everything, so what are you going to prioritize? As an example, if your system depends on an external library, you can explicitly state that you will assume that external library has already been verified by its implementation team. —SS]

2.3 Custom Documents

[Summarize the CDs (Custom Documents) that were tackled by this project. CDs can include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. CDs should have already been approved by the course instructor as included in your problem statement. You can use a pull request to update your CDs (in TeamComposition.csv or Repos.csv) if your plan changes as a result of the VnV planning exercise. —SS]

2.4 Relevant Documentation

[Reference relevant documentation. This will definitely include your SRS and your other project documents (design documents, like MG, MIS, etc). You can include these even before they are written, since by the time the project is done, they will be written. You can create BibTeX entries for your documents and within those entries include a hyperlink to the documents. —SS]

Citation Example of SRS[1]

[Don't just list the other documents. You should explain why they are relevant and how they relate to your VnV efforts. —SS]

3 Plan

[Introduce this section. You can provide a roadmap of the sections to come. —SS]

3.1 Verification and Validation Team

[Your teammates. Maybe your supervisor. You should do more than list names. You should say what each person's role is for the project's verification. A table is a good way to summarize this information. —SS]

3.2 SRS Verification

[List any approaches you intend to use for SRS verification. This may include ad hoc feedback from reviewers, like your classmates (like your primary reviewer), or you may plan for something more rigorous/systematic. —SS]

[If you have a supervisor for the project, you shouldn't just say they will read over the SRS. You should explain your structured approach to the review. Will you have a meeting? What will you present? What questions will you ask? Will you give them instructions for a task-based inspection? Will you use your issue tracker? —SS]

[Maybe create an SRS checklist? —SS]

3.3 Design Verification

[Plans for design verification —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.4 Verification and Validation Plan Verification

[The verification and validation plan is an artifact that should also be verified. Techniques for this include review and mutation testing. —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.5 Implementation Verification

[You should at least point to the tests listed in this document and the unit testing plan. —SS]

[In this section you would also give any details of any plans for static verification of the implementation. Potential techniques include code walkthroughs, code inspection, static analyzers, etc. —SS]

[The final class presentation in CAS 741 could be used as a code walkthrough. There is also a possibility of using the final presentation (in CAS741) for a partial usability survey. —SS]

3.6 Automated Testing and Verification Tools

[What tools are you using for automated testing. Likely a unit testing framework and maybe a profiling tool, like ValGrind. Other possible tools include a static analyzer, make, continuous integration tools, test coverage tools, etc. Explain your plans for summarizing code coverage metrics. Linters are another important class of tools. For the programming language you select, you should look at the available linters. There may also be tools that verify that coding standards have been respected, like flake9 for Python. —SS]

[If you have already done this in the development plan, you can point to that document. —SS]

[The details of this section will likely evolve as you get closer to the implementation. —SS]

3.7 Software Validation

[If there is any external data that can be used for validation, you should point to it here. If there are no plans for validation, you should state that here. —SS]

[You might want to use review sessions with the stakeholder to check that the requirements document captures the right requirements. Maybe task based inspection? —SS]

[For those capstone teams with an external supervisor, the Rev 0 demo should be used as an opportunity to validate the requirements. You should plan on demonstrating your project to your supervisor shortly after the scheduled Rev 0 demo. The feedback from your supervisor will be very useful for improving your project. —SS]

[For teams without an external supervisor, user testing can serve the same purpose as a Rev 0 demo for the supervisor. —SS]

[This section might reference back to the SRS verification section. —SS]

4 System Tests

[There should be text between all headings, even if it is just a roadmap of the contents of the subsections. —SS]

4.1 Tests for Functional Requirements

[Subsets of the tests may be in related, so this section is divided into different areas. If there are no identifiable subsets for the tests, this level of document structure can be removed. —SS]

[Include a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. —SS]

4.1.1 Area of Testing1

[It would be nice to have a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. If a section covers tests for input constraints, you should reference the data constraints table in the SRS. —SS]

Title for Test

1. test-id1

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs. Output is not how you are going to return the results of the test. The output is the expected result. —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

2. test-id2

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

4.1.2 User Interface Tests for Functional Requirements

The system tests listed below are planned to verify the implemented system as a whole against functional requirements related to the user interface component of the system. These requirements can be found in the SRS, under section S.2's functional requirements for the 'User Interface System' (PUT REFERENCE HERE).

1. **SFT-2.1** Name

   Control: Manual

Initial State: Initial State

Input: Input

Output: Output

Test Case Derivation: Test Case Derivation

How test will be performed: How test will be performed

4.2 Tests for Nonfunctional Requirements

[The nonfunctional requirements for accuracy will likely just reference the appropriate functional tests from above. The test cases should mention reporting the relative error for these tests. Not all projects will necessarily have nonfunctional requirements related to accuracy. —SS]

[For some nonfunctional tests, you won't be setting a target threshold for passing the test, but rather describing the experiment you will do to measure the quality for different inputs. For instance, you could measure speed versus the problem size. The output of the test isn't pass/fail, but rather a summary table or graph. —SS]

[Tests related to usability could include conducting a usability test and survey. The survey will be in the Appendix. —SS]

[Static tests, review, inspections, and walkthroughs, will not follow the format for the tests given below. —SS]

[If you introduce static tests in your plan, you need to provide details. How will they be done? In cases like code (or document) walkthroughs, who will be involved? Be specific. —SS]

4.2.1 Area of Testing1

Title for Test

1. test-id1

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input/Condition:

Output/Result:

How test will be performed:

2. test-id2

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input:

   Output:

   How test will be performed:

4.2.2 Area of Testing2

...

4.3 Traceability Between Test Cases and Requirements

Table 1: RealmForge System Test Requirements Traceability Summary

System Test ID	Requirements Coverage	Related Requirements
SFR-001	ENV-002	ENV-003
SFT-2.1	SF4-404	SF4-404

5 Unit Test Description

TBD

References

[1] Aeolus Software (https://github.com/Aeolus-Software), "Realmforge system requirements specification," https://github.com/Aeolus-Software/RealmForge/blob/main/pdfs/SRS/SRS.pdf, Aeolus Software, 2026.

6 Appendix

This is where you can place additional information.

6.1 Symbolic Parameters

The following symbolic constants are used throughout this document:

Table 2: Symbolic Constants Used in the Verification and Validation Plan

Constant	Value	Unit
CONSTANT_EXAMPLE_1	1	s

6.2 Usability Survey Questions?

[This is a section that would be appropriate for some projects. —SS]

7 Appendix — Generative AI Use Disclosure

7.1 AI Tools Used

- Anthropic Claude Sonnet 5.5
- Google Search AI Overview (Google Gemini)

7.2 How and Where Used

AI tools were used for assistance with LaTeX syntax and commands, during the development of sections SRS Verification and Design Verification, and when developing LaTeX formatting for the document as a whole. Specifically, AI tools were used for assistance in creating the LaTeX custom system test format macro used throughout the document, as well as the symbolic parameter table used in section 6.1. In these sections, AI tools were not used to generate any content text (LaTeX formatting code was generated), or wholly generate any of the ideas or conclusions presented.

7.3 Verification of AI-Generated Content

Any LaTeX syntax or code provided by AI tools was verified by testing compilation in LaTeX documents, and ensuring the resulting output was correctly formatted and matched the needs of the document. No ideas or text (outside of LaTeX formatting code) were newly generated by AI tools and presented in the document.

8 Appendix — Reflection

Reflection Questions Answered:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. What knowledge and skills will the team collectively need to acquire to successfully complete the verification and validation of your project? Examples of possible knowledge and skills include dynamic testing knowledge, static testing knowledge, specific tool usage, Valgrind etc. You should look to identify at least one item for each team member.
4. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Andreas:

1.
2.
3.
4.

Humna:

1.
2.
3.
4.

Christopher:

1.

2.

3.

4.

Owen:

1.

2.

3.

4.

Andy:

1.

2.

3.

4.